



The Sugar Trap — Market Gap Analysis

Client: Helix CPG Partners Analyst: Lawrence Koomson | BSc. IT — Data Science & Analytics University of Cape Coast | AmaliTech Capstone 2026



The Business Problem

Helix CPG Partners advises major food manufacturers on new product development.

Their client — a global snack manufacturer — wants to launch a "Healthy Snacking" line.

The Key Question: 🔍 "Where is the Blue Ocean in the snack aisle?"

They need data to prove WHERE the specific market gaps are — areas where consumer demand for High Protein + Low Sugar products is NOT being met.



Data & Methodology



Dataset: Open Food Facts — World's largest open food database



Raw Data: 500,000 rows loaded from CSV



After Cleaning: 109,564 high-quality products



Categories Created: 8 high-level product buckets

Cleaning Steps: • Removed missing product names • Removed rows missing sugar AND protein data • Filtered biologically impossible values (>100g per 100g) • Removed products where combined macros exceeded 105g

The Nutrient Matrix — Finding the Blue Ocean



Sugar threshold: 14.0g | Protein threshold: 10.3g per 100g



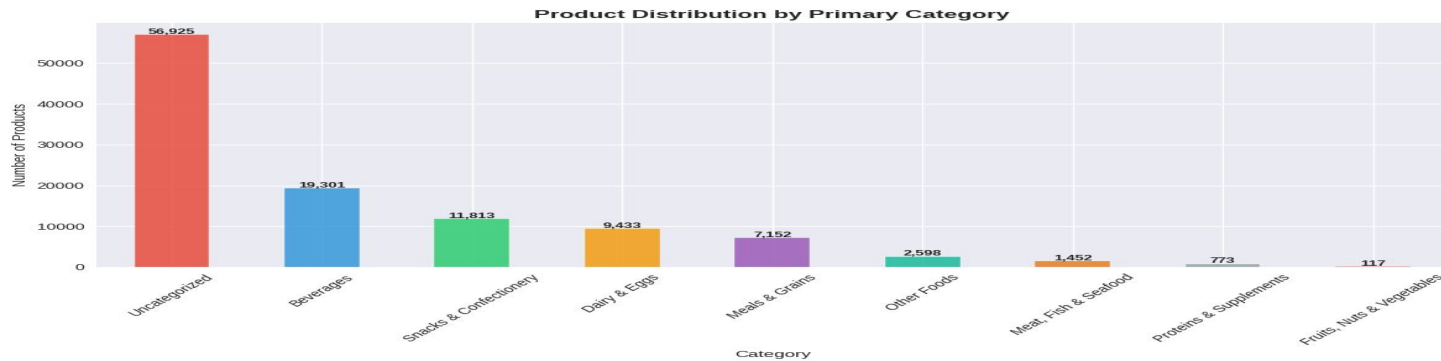
Key Findings

 **Blue Ocean (High Protein + Low Sugar)** • 11,869 products in this underserved space • Average protein: 18.7g per 100g • Average sugar: 2.8g per 100g

 **Oversaturated Zone (High Sugar + Low Protein)** • 11,946 products crowding this space • 54.1% score E (Worst) on Nutri-Score • Dominated by Snacks & Confectionery

 **Health Validation** • Blue Ocean products: 24.5% score A (Best) on Nutri-Score • Genuinely healthier — not just low-sugar coincidentally

Category Analysis



Beverages and Snacks dominate the market. Dairy & Eggs represents the strongest Blue Ocean opportunity.



The Recommendation for Helix CPG Partners

KEY INSIGHT:

*"Based on the data, the biggest market opportunity is in **Dairy & Eggs**, specifically targeting products with **20.2g of protein** and less than **4.3g of sugar**."*

 **Top Protein Sources for R&D Team:** • #1 Soy/Soya — 2,943 products (30.3%) • #2 Peanuts — 938 products (9.7%)
• #3 Whey — 868 products (9.0%)

 **Candidate's Choice — Nutri-Score Validation:** Blue Ocean products score A/B (healthy) while 54.1% of the Oversaturated zone scores E (Worst). This confirms the gap is REAL and worth investing in.



Thank You

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Data tells the story — we just need to listen.